ADITYA VERMA

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EXPERIENCE

Software Engineer, Tata Consultancy Services - *Mumbai, Maharashtra* November 2025- Current

- Engineered backend Python and PySpark services for the Reserve Bank of India's CIMS modernization, migrating 230+ regulatory returns from a report-based repository to an element-based architecture spanning 50,000+ elements and serving filings from 5,000+ banks and financial institutions.
- Cut end-to-end batch runtime from 60 to 15 minutes by re-tuning PySpark resource allocation and refactoring execution logic, raising per-run throughput from ~50 to 200+ elements while eliminating timeouts and freeing contended cluster resources for concurrent workloads.
- Built dual ingestion flows handling both direct element-based collection and report-to-element conversion, implementing dimension-mapping logic that guarantees lossless transformation across bulk historical migration and daily production batch runs processing returns of 20,000–100,000+ rows each.
- Designed automated validation and sanity-check routines gating every environment promotion, catching defects pre-release and driving 20 returns to live production status for active bank filing.
- Lead a team of 3 engineers in a fast-paced Agile environment, coordinating with cross-functional teams on requirements, defect triage, and release readiness across legacy-to-new system migrations.

EDUCATION

B.Tech in Computer Science and Engineering - Graphic Era University, Dehradun | CGPA: 8.32 **2021 - 2025**

PROJECTS

Sales Pipeline Revenue Risk & Growth Action Copilot *(TCS AI Hackathon - National Finals, September, 2026)* :

- Built a multi-source intelligence layer that sits above existing GTM systems rather than replacing them, ingesting and correlating signals from CRM, ticketing, internal messaging, and email to surface revenue risks invisible when CRM data is treated as the sole source of truth. Designed three role-scoped interfaces over a shared signal pipeline: account executive, RevOps, and VP/CRO, each exposing a different decision surface, from next-best-action recommendations and stage-hygiene prompts to approval workflows and portfolio-level forecasting.
- Implemented deterministic risk and growth scoring that updates in real time as new signals arrive, with per-signal attribution showing which inputs moved each score, paired with XGBoost models for revenue forecasting and win probability.
- Engineered a retrieval-augmented reasoning layer (GPT-4o-mini and Claude Sonnet 4.6) with gated invocation that resolves cases deterministically wherever possible and routes to an LLM only when genuinely required, cutting token spend while keeping outputs grounded in the underlying signal evidence.
- Architected the system as human-in-the-loop by design, restricting AI to recommendations while all state changes remain under user control, delivered as a production-grade application in 5 days on a Python, React, and PostgreSQL stack, with Kafka and Redis being added for horizontal scale.

Self-Driving Car Agent (September, 2023 - December, 2023) : github.com/addy-301/Self-Driving-Car-Agent-with-RL

- Built an autonomous driving agent in Unity3D with ML-Agents, designing a custom simulation track and implementing vehicle control, sensor integration, and reward-shaping logic in C# to train a driving policy end-to-end.
- Trained and benchmarked PPO and SAC over 800k steps, quantifying the sample-efficiency vs. training-cost trade-off: SAC converged to ~50% higher cumulative reward (144 vs 94) while PPO reached a comparable policy in 33 minutes of wall-clock training against SAC's 2.5 hours; diagnosed SAC's entropy collapse and PPO's sustained exploration from TensorBoard policy-entropy and loss curves.

SKILLS

Programming Languages: Python, Java, TypeScript, HTML, CSS, C++
Frameworks: React.js, Node.js, Express.js, REST APIs, Tailwind CSS, PySpark
Database & Backend: SQL, MongoDB, Mongoose
AI / Machine Learning: Generative AI, Retrieval-Augmented Generation (RAG), Agentic AI, Reinforcement Learning
Cloud & Tools: Microsoft Azure, Google Cloud Platform (GCP), Git, GitHub, Docker, Vercel
Core Concepts: SDLC, Agile/Scrum, Object-Oriented Programming (OOP), Data Structures and Algorithms

ACHIEVEMENTS & CERTIFICATIONS

- **Innovation Superstar Award, TCS** - Recognized by project leadership for sustained performance and for being the only engineer from the business unit to win the Mumbai Regional Finals and represent the region at the TCS AI Hackathon National Finals.
- **TCS AI Hackathon - National Finalist:** One of 32 teams to reach the National Finals from 22,000+ participants, winning the Mumbai Regional Finals. Designed and shipped a production-grade AI application in 5 days and presented to TCS executive leadership, including the CIO and business unit heads.
- **Intra-University Coding Competition** - Secured 3rd rank in a coding contest hosted by Graphic Era University.

Certifications: Claude Certified Developer - Foundations (Anthropic) · Salesforce Agentblazer Innovator (Salesforce) · AWS GenAI Practitioner (AWS) · Introduction to Generative AI (Google Cloud) · Google Cloud Computing Fundamentals (Google Cloud)